Scheduling

Introduction to Scheduling Process Behavior

CPU bound vs. I/O bound processes

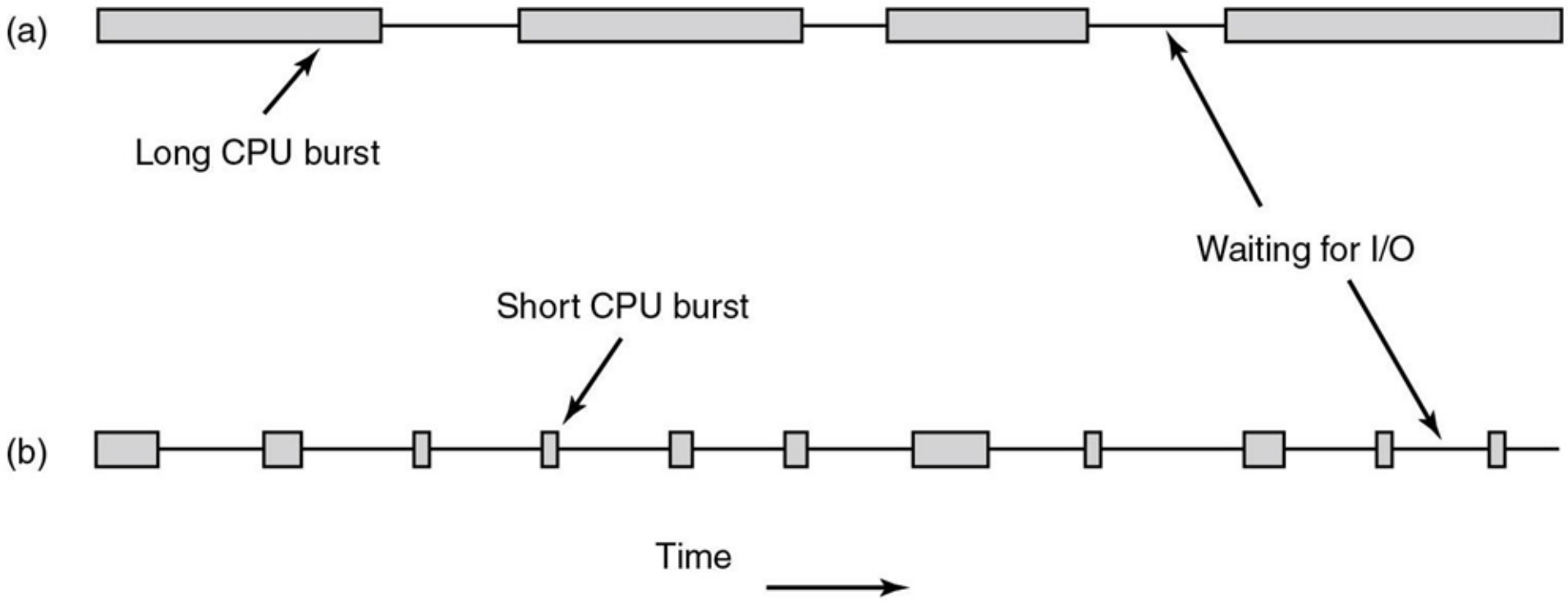
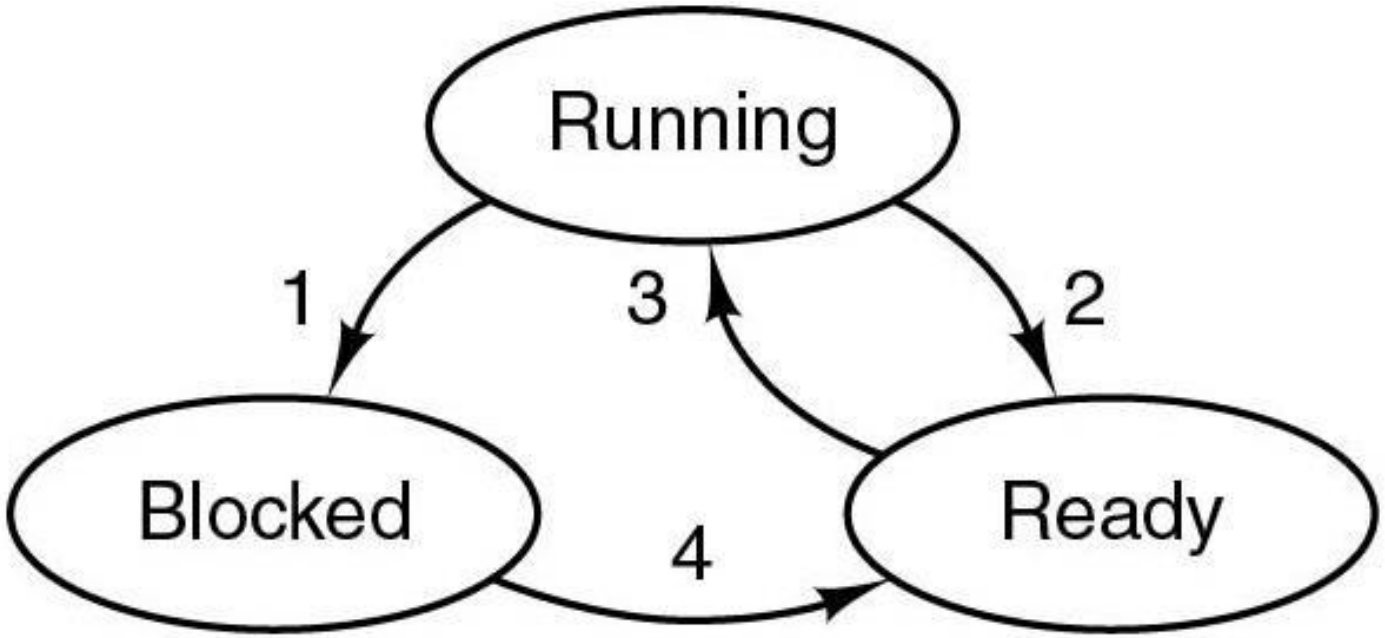


Figure 2.40 Bursts of CPU usage alternate with periods of waiting for I/O. (a) A CPU-bound process. (b) An I/O-bound process.

Process State Revisited



If more processes ready than CPU s available:

- **Scheduler** decides which process to run next
- Algorithm used by scheduler is called **scheduling algorithm**

When to Schedule?

- ***Process exits***
- ***Process blocks on I/O, Semaphore, etc.***
- *When a new process is created*
- *When an interrupt occurs:*
  - *I/O, clock, syscall, etc.*

Preemptive vs **non-preemptive** scheduling?

Categories of Scheduling Algorithms

- Batch System - non-preemptive scheduling
- Interactive System - preemptive scheduling
- Real Time System - different category

Scheduling Algorithm Goals (1 of 2)

- Different goals for different systems
  - Batch
  - Interactive
  - Real time
- All systems:
  - Fairness - giving each process a fair share of the CPU
  - Policy enforcement - seeing that stated policy is carried out
  - Balance - keeping all parts of the system busy

Batch Systems

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- Throughput : maximize jobs per hour
- Turnaround time : minimize time between submission and termination
- CPU utilization : keeping the CPU busy all the time

First-Come First-Served

- Process jobs in order of their arrival
- Non-preemptive
- Single Process Queue
  - New jobs or blocking processes are added to the end of the queue
- “Convoy Effect” if only few CPU bound and many I/O bound processes

Shortest Job First (1 of 2)

- Pick the job with the shortest run time
- Provably optimal:
  - Lowest turnaround time

Shortest Job First (2 of 2)

- Pick the job with the shortest run time
- Provably optimal:
  - Lowest turnaround time
  - Only if all jobs are available simultaneously
  - If new jobs arrive, it may lead to starvation
- Runtimes have to be known in advance
- Highest-response-ratio-next
  - Improved version of Shortest Job First

Interactive Systems

- **Response time:** respond to requests quickly
- **Proportionality:** meet users' expectations

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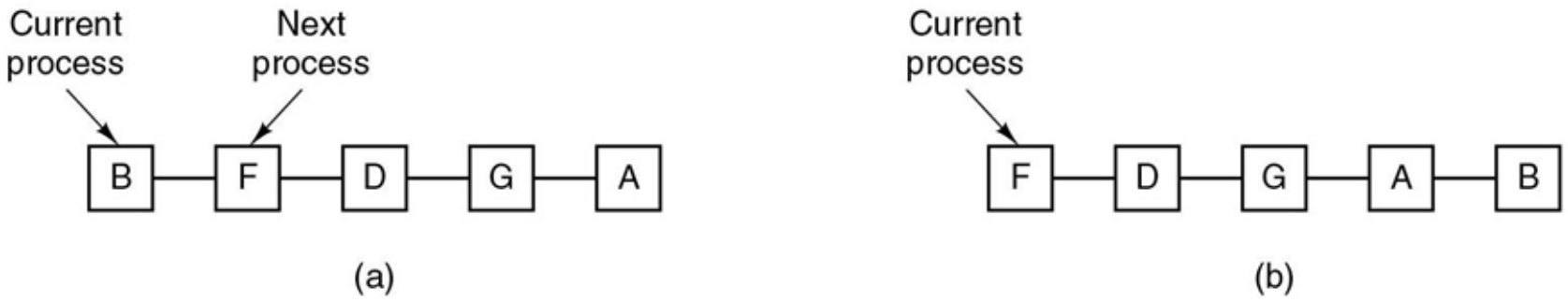
Apple MacIntosh (1984)

Round Robin Scheduling

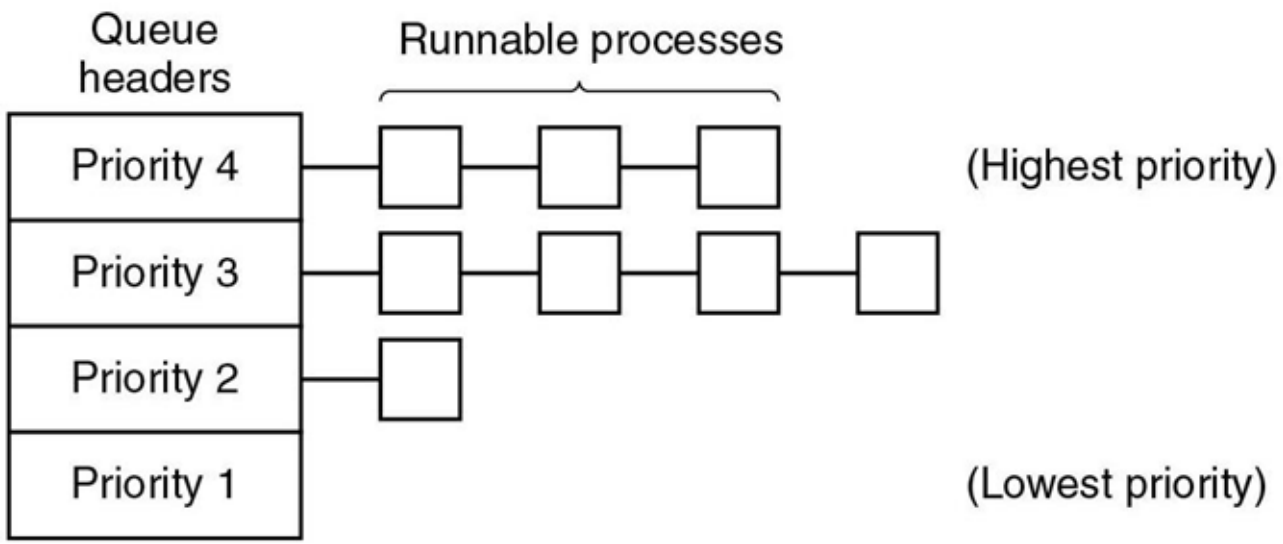
- Preemptive scheduling algorithm
- Each process gets a **time slice** or **quantum**
- If process is still running at end of quantum it gets preempted end goes to end of ready queue
- Question: How big should the quantum be?

CPU utilization vs. response time



Priority Scheduling (1 of 2)

Simplest, multiple queues



- Similar to round robin but several ready queues
- Next process is picked from queue with highest priority
- Static vs. dynamic priorities
- What processes should have high priorities?

Priorities Are Complicated

and may not work out the way you think...

(remember priority inversion?)

Shortest Process Next

Problem: How to minimize response time for each priority queue?

Idea: Use shortest "job" first and try to best predict next running time

Solution: Form weighted average of previous running times of process $\rightarrow$ **Aging**

$$T_{k+1} = a * T_{k-1} + (1 - a) * T_k$$

Easy to implement when $a = 1/2$

Guaranteed Scheduling

Idea: N processes running $\longrightarrow$ each process gets 1/Nth of CPU time
(also known as fair-share)

Solution:

- Calculate how much C P U time it might have gotten:
  - Time since process creation divided by N
- Measure actual consumed C P U time and form ratio
- 0.5 $\longrightarrow$ process running half the time it was entitled to
- 2.0 $\longrightarrow$ process running twice as much as it was entitled to
- Pick process with the smallest ratio to run next
- Note: fair-share scheduling can be per-user/-process

Lottery Scheduling

- Processes get lottery tickets
- Whenever a scheduling decision is made O S chooses a winning ticket randomly

- Processes can possess multiple tickets $\longrightarrow$ Priorities

- Tickets can be traded between processes.
- Tickets are immediately available to newly created processes.

Policy vs. Mechanism

Important principle

Here: we may have a scheduling algorithm, but parameters to be filled in by user (process)

For instance, to give some child processes higher priority than others

Real Time Systems (1 of 3)

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N A S A

- Meeting deadlines: avoid losing data
- Predictability: avoid quality degradation in multimedia systems

Real Time Systems (2 of 3)

- Systems where timing plays essential role
- Soft real time vs. Hard real time
- Can consist of periodic and aperiodic tasks
- Schedules can be static or dynamic
- System with periodic tasks is schedulable when:

$$\sum_{i=1}^{m} \frac{C_i}{P_i} \leq 1$$

Real Time Systems (3 of 3)

Consider three jobs:

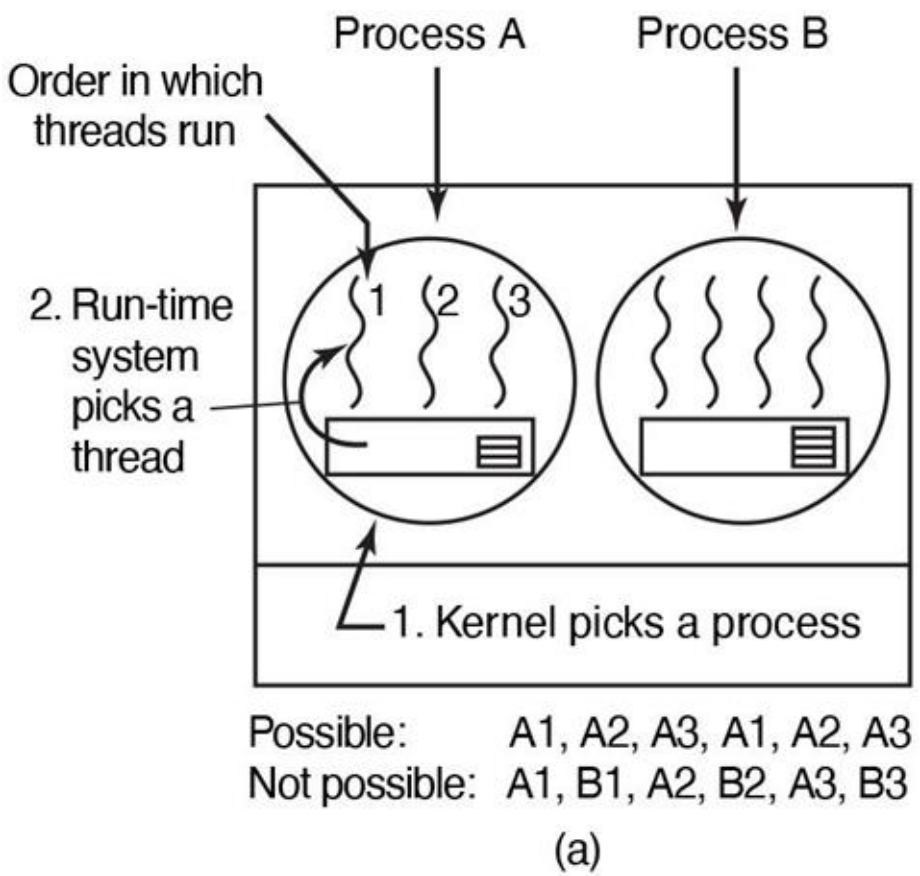
-	**Period**	**Req. C P U time**
P1	100 ms	50 ms
P2	200 ms	30 ms
P3	500 ms	100 ms

Is this system schedulable?

Scheduling Algorithm Goals (2 of 2)

- Interactive systems
  - Response time - respond to requests quickly
  - Proportionality - meet users' expectations
- Real-time systems
  - Meeting deadlines - avoid losing data
  - Predictability - avoid quality degradation in multimedia systems

Thread Scheduling (1)



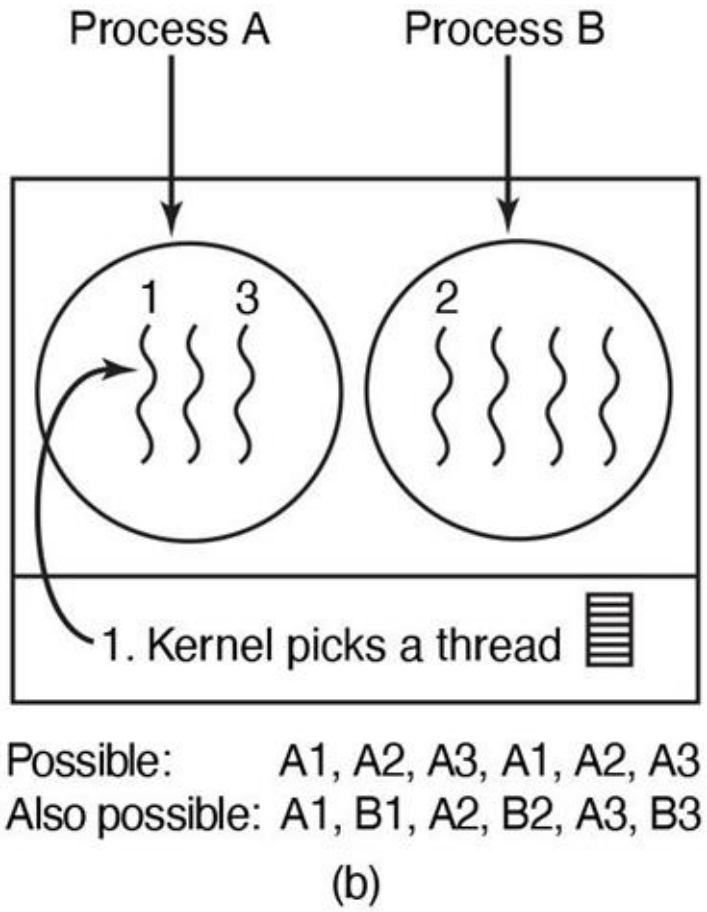


Figure 2.45. (a) Possible scheduling of user-level threads with a 50-msec process quantum and threads that run 5 msec per CPU burst.

Figure 2.45 (b) Possible scheduling of kernel-level threads with the same characteristics as (a).

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